

Phase Checkpoint — FastAPI Simulation API & Docker Containerization

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1 Session Objectives

1. Add `dry_run` flag to `EnrichmentCore` for read-only simulation mode
2. Fix missing SHAP explanations in `DecisionCore` output (bug affecting streaming pipeline)
3. Validate refactored cores via updated E2E test script
4. Build FastAPI simulation API (`src/api/`)
5. Containerize simulation API with Docker Compose

2 Bug Fix — Missing SHAP Explanations

Root cause: `DecisionCore._build_output()` attached 7 identifier fields from the scored event dict but did not carry through `ae_explanation` or `lstm_explanation`. These fields existed in the scored dict (placed there by `ScoringCore`) and were passed *into* `DecisionInput`, but the `Decision.to_dict()` output did not include them, and `_build_output` did not re-attach them.

Impact: The `decisions` RedPanda topic was missing SHAP explanations for all events — affecting both the streaming pipeline and any downstream consumer (dashboard, audit log).

Fix: Two lines added to `_build_output`:

```
output["ae_explanation"] = scored.get("ae_explanation")
output["lstm_explanation"] = scored.get("lstm_explanation")
```

3 Design Decision — Simulation Dry-Run

Problem: Simulation mode must run the full `Enrichment` → `Scoring` → `Decision` pipeline without contaminating real profiles, buffers, or sequence data in Redis.

Analysis: Only `EnrichmentCore` has write side effects (`_update_buffers`, `_update_sequence`). `ScoringCore` and `DecisionCore` are already read-only.

Options evaluated:

- A. `dry_run=False` parameter — streaming path unchanged, simulation opts in
- B. Fake Redis wrapper that drops writes — over-engineered, fragile
- C. Split into `compute` + `commit` phases — clean but risky (forgotten `commit()` silently breaks streaming)

Decision: Option A. Safe by default — streaming path never changes, simulation explicitly requests read-only mode. Reliability over elegance.

```
def enrich_event(self, event: dict, dry_run: bool = False) -> dict:
    # ... feature computation ...
    if not dry_run:
        self._update_buffers(...)
        self._update_sequence(...)
    return features
```

4 E2E Pipeline Validation

Test script rewritten to use refactored cores (previously used `EnrichmentService.__new__()` hack to bypass streaming wrapper).

```
# Composition root (same pattern as FastAPI lifespan)
enrichment_core = EnrichmentCore(redis_client=r, models_dir=...)
scoring_core = ScoringCore(redis_client=r, fusion=fusion, ...)
decision_core = DecisionCore(redis_client=r, engine=engine)

# Pipeline: one method call per stage
enriched = enrichment_core.enrich_event(event)
scored = scoring_core.score_event(enriched)
decision = decision_core.decide_event(scored)
```

Results (500 events, last 20 evaluated):

Metric	Value
Score range	0.0024 — 1.0000
Mean score	0.4423
Above REVIEW threshold	4 / 20
Tier distribution	14 INFO / 3 REVIEW / 3 BLOCK
SHAP explanations present	Yes (AE confirmed)

Results match pre-refactor baseline. Core extraction validated.

5 FastAPI Simulation API — `src/api/`

5.1 Architecture

File	Responsibility	Pattern
<code>models.py</code>	Pydantic request/response schemas	Cross-field validation
<code>routes.py</code>	Endpoint handlers + event builder	Single responsibility
<code>app.py</code>	Lifespan DI wiring, router mount	Composition root

Key design decisions:

- **Separate from streaming health service** — different responsibility, different failure domain, independent deployability
- `app.state` over `Depends()` — cores are singletons created once at startup, not per-request objects
- `/api/v1/` **prefix** — versioned from the start

5.2 Request Model

```
SimulationRequest {
  client_id : str (supervisor provides)
  employee_id : str (supervisor provides)
  amount : float (supervisor provides)
  operation_type : OperationType enum [retrait|versement|virement|cheque]
  payload : Optional[Union[VirementPayload, ChequePayload]]
}

VirementPayload { beneficiary_id : str }
ChequePayload { emitter_id : str }
```

Cross-field validation via `@model_validator`: `virement` → requires `VirementPayload`, `cheque` → requires `ChequePayload`, `retrait/versement` → payload must be `None`.

5.3 Auto-Generated Fields

Field	Source
<code>event_id</code>	<code>uuid4()</code>
<code>timestamp</code>	<code>datetime.now()</code> (tz-naive, matches system)
<code>account_id</code>	"N/A" (not used by any core)
<code>branch_id</code>	"SIM-000"
<code>currency</code>	"TND"
<code>channel</code>	"guichet"

5.4 Response Model

```
SimulationResponse {
  // Identifiers
  event_id, client_id, account_id, employee_id,
  branch_id, timestamp, amount, operation_type

  // Decision
  tier : str
  fused_score : float
  reasons : list[str]
  regulatory_flags : list[str]

  // Explainability
  ae_explanation : Optional[dict]
  lstm_explanation : Optional[dict]
}
```

6 Docker Containerization

6.1 Dockerfile.api

- Base image: `python:3.11-slim`
- Layer-optimized: `requirements.txt` copied and installed first (cached on rebuild), `src/` copied second (changes frequently)
- CPU-only PyTorch via `--extra-index-url https://download.pytorch.org/whl/cpu`
- Models excluded from image — mounted as volume at runtime

6.2 Docker Compose Services

Service	Image	Port	Notes
redpanda	redpandadata/redpanda:latest	9092	Kafka-compatible broker
redis	redis:7-alpine	6379	Hot profiles, buffers, sequences
simulation-api	Custom build	8000	FastAPI simulation endpoint

Key configuration:

- `REDIS_HOST=redis` — container-to-container via service name
- `depends_on: redis` — startup ordering
- `volumes: ./models:/app/models` — models mounted at runtime, supports weekly retrain without image rebuild

7 Test Results — Containerized API

Test 1: 15,000 DT versement (salaried client)

Field	Value
Tier	BLOCK
Fused score	0.999803
Regulatory flags	Amount 15000.00 DT $\geq$ 10000 DT reporting threshold (BCT)
AE top features	<code>is_round_amount</code> , <code>arch_salaried</code> , <code>amount_to_mean_ratio</code> (5.05x)

Test 2: 9,500 DT virement to new beneficiary (same client)

Field	Value
Tier	BLOCK
Fused score	0.999902
Regulatory flags	Virement of 9500.00 DT to new beneficiary
AE top features	<code>amount_to_mean_ratio</code> (19.3x), <code>z_amount</code> (11.6), <code>arch_salaried</code>

Both tests produced identical results locally and in Docker container.

8 Next Session

1. Health endpoints for streaming consumers
2. Streamlit dashboard (consumes decisions topic)
3. MLflow + Airflow integration
4. Prometheus/Grafana observability